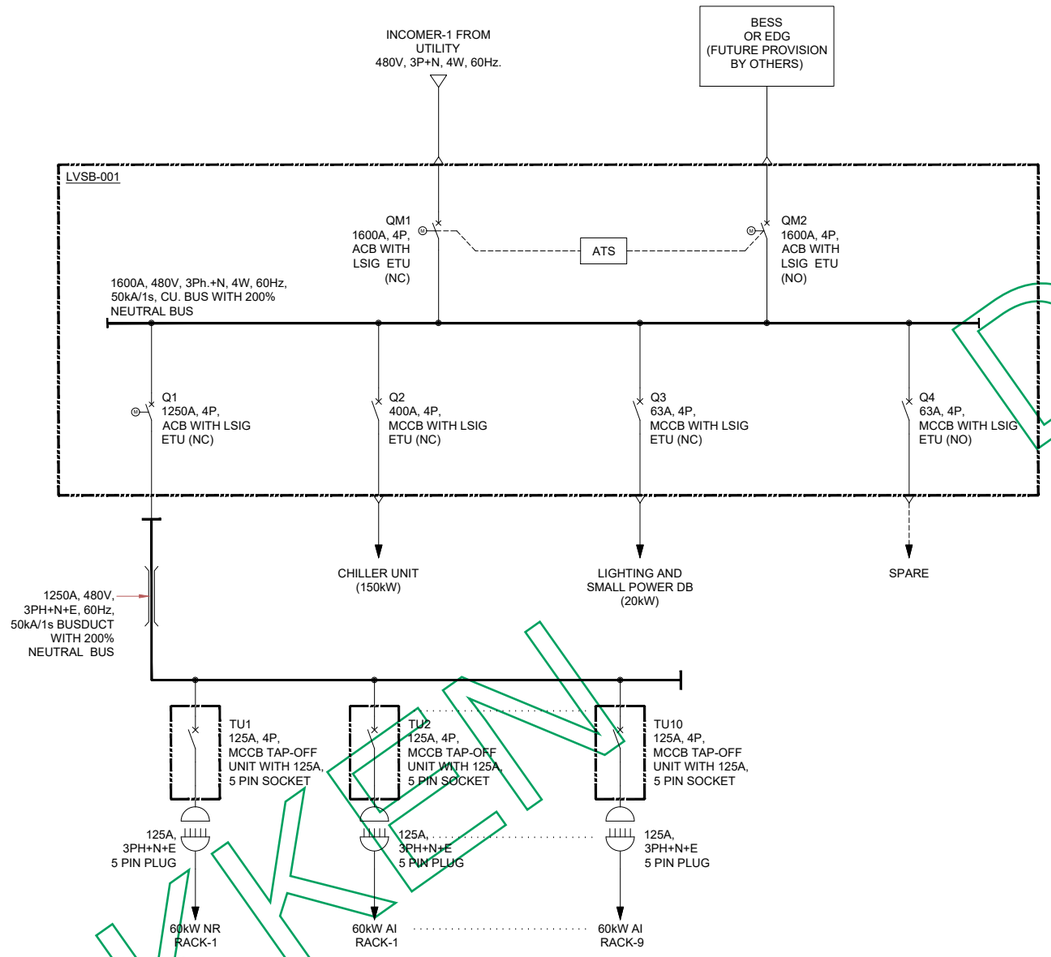

TYPICAL SLD FOR ONE CONTAINERIZED
DATA CENTER MODULE

NOTES:

1. The busbars for the LV Switchgear and associated Busway system shall be constructed from 99.9% pure high-conductivity Electrolytic Copper. The neutral busbar for all 480V LV Switchgear sections and Busway runs feeding IT or rack loads shall be rated at 200% of the phase busbar cross-sectional area, to accommodate potential harmonic currents arising from non-linear IT equipment.
2. LV switchboard to comply with NMX-J-515-ANCE-2021 & IEC 61439-2 Form 2b separation – busbars isolated from functional units and terminals. IEC definition applies where local standard is silent.
3. All busbars main and branch shall be fully insulated. Joints shall be covered with insulating shrouds.
4. Switchgear, Busduct and associated equipment shall be suitable for a supply voltage variation of ±10% and a frequency variation of ±5%.
5. Design ambient temperature for the switchgear and busduct shall be 40°C. However, all equipment shall be suitable for normal operation at 50°C for at least 2 hours.
6. The ratings of the circuit breakers mentioned in the SLD shall be the in-panel rating of the circuit breakers without any derating.
7. The rating of the LV busduct mentioned in the RFQ shall be the net site rating at the design ambient temperature of 40°C without any derating.
8. The vendor shall furnish relevant data, test certificates and details of past orders of similar equipment for the specified ambient conditions.
9. 480V LV Switchboard to be type-tested per IEC 61439-1/2 & NMX-J-515-ANCE-2021, rated 50 kA/1s, with internal arc classification to IEC TR 61641 (criteria 1-7) and Form 2b separation. Vendor to submit accredited type-test certificates with bid.
10. 480 V LV Bus Duct to be type-tested per IEC 61439-6 & NMX-J-515-ANCE-2021, rated 50 kA/1 s. Vendor shall submit accredited type-test certificates with bid.
11. Switchgear shall be suitable for bottom cable entry for cable termination access from the front.
12. The busduct connection between the switchgear and the data hall distribution shall be routed from the top of the switchgear assembly. The busway system shall be equipped with 10 plug-in tap-off units for rack power distribution.
13. The busway shall be a 5-wire system (3Ph + N + PE), with the neutral busbar rated at 200% of the phase conductor cross-section to accommodate harmonic currents from IT loads.
14. Each plug-in tap-off unit of the busway shall be equipped with the following components:
    - One (1) 480V, 125 A, 4-pole, 60 Hz MCCB with LSIG electronic trip unit, providing overload, short-circuit, and earth-fault protection.
    - One (1) 480V, 125 A, 5-pin (3Ph + N + E), 60 Hz industrial socket outlet, suitable for flexible whip connection to the rack power distribution unit (PDU).
15. Incomer Breakers shall be provided with LSIG Electronic trip unit with Modbus TCP/IP communication ports for remote SCADA communication.
16. Multifunction meters shall be provided on all Incomers and Outgoing feeders. All MFMs shall have communication ports for communicating with higher level SCADA system over Modbus TCP/IP protocol.
17. Network switch (managed 2-layer type) shall be provided inside the switchgear. All protection relays and MFMs shall be connected to this network switch for further connection to higher level SCADA system for monitoring and control.
18. Undrilled, removable, non-magnetic gland plates minimum 4mm thick shall be provided on all switchgears.
19. LV Switchgear shall have facilities for extension at both ends.
20. Panel space heaters shall be provided in LV Switchboard which shall be fed from 277V, 1 ph, AC space heater bus (space heater bus fed from external essential power supply). Heaters shall be protected with MCB + RCD (30mA sensitivity). When heating system is live, this shall be indicated by means of prominently situated red LED.
21. CTs for metering and protection shall be with 1A secondary. CT class for metering shall be Class 0.5 and for protection shall be Class 5P20 as a minimum. Detailed CT/VT sizing calculation shall be submitted by Vendor during detail engineering.
22. The neutral shall be switched. Neutral links shall not be used. All MCBs used for auxiliary supplies shall be 2P/4P, C curve type.
23. Final paint finish for the Switchgear and Bus duct enclosure shall be RAL 7035.
24. All Incoming and outgoing ACBs shall be motorized type with LSIG ETU having Modbus TCP/IP communication ports.
25. All outgoing feeder MCCBs shall be provided with LSIG ETU.

LOAD DETAILS PER DATA CENTER MODULE

2 NETWORK RACKS (NR) AND 18 AI RACKS (EACH RATED 60kW)	1200kW
CHILLER UNIT (2 NO. OF 150kW)	300kW
LIGHTING & SMALL POWER	20kW
TOTAL LOAD PER DATA CENTER MODULE	1520kW
TOTAL NO. OF SWITCHGEAR	2 Nos.
LOAD PER SWITCHGEAR	760kW

REV	DATE	DETAILS	BY	CHK	APP
A	04/11/25	ISSUED FOR PROPOSAL	AM	VA	MB

CLIENT: Crusader Project Services L.L.C-FZ

DRAKKEN

© Design Rights DRAKKEN. This document and the information it contains is confidential and the property of DRAKKEN and is not to be reproduced, communicated to a third party or used in any manner whatsoever without the prior written consent of DRAKKEN.

TITLE:
CRUSADER MEXICO CITY AI FACTORY
POWER DISTRIBUTION SINGLE LINE DIAGRAM

DRAKKEN DRAWING NO. 250134-PRO001-(U)SL-001

FORMAT:A1	DRAWING NO.	REV.	SHT. NO.
SCALE:NTS	-	A	1/1

DRAKKEN